Problem Statement

Why does BIS compliance remain out of reach for Indian MSEs?

Indian Micro and Small Enterprises (MSEs) manufacturing building materials are legally required to conform to Bureau of Indian Standards (BIS) specifications. Yet identifying the correct standard remains the biggest barrier — not the compliance itself.

570+

Standards to Search

The BIS SP-21 document spans 929 pages and 570+ building material standards across 27 sections — cement, steel, aggregates, concrete and more. No quick-search tool exists for non-experts.

2–4 wks

Compliance Discovery Delay

Without automation, engineers and business owners spend 2–4 weeks manually cross-referencing product specs with standard titles, descriptions and scope clauses.

High Risk

Cost of Non-Compliance

Selecting the wrong or outdated standard leads to BIS licence rejection, product recall, or legal penalties — disproportionately harming micro enterprises with limited legal support.

Solution Overview

From product description to BIS standard — in under 2 seconds

570 BIS Standards Indexed

Full SP-21 Building Materials document (929 pages) parsed into 570 structured chunks — one per standard, with category tags.

Hybrid Retrieval (BM25 + Embeddings)

BM25 keyword scorer (40%) fused with all-MiniLM-L6-v2 semantic embeddings (60%) for high-accuracy recommendations.

LLM-Powered Query Understanding

A language model interprets user queries — resolving synonyms, domain terms and ambiguous product descriptions before retrieval.

Fully Offline & Fast

Runs on CPU with no external API dependency. Average query latency is 1.3 seconds — well within the 5-second target.

Graceful Fallback

Auto-switches to BM25-only mode if the embedding model is unavailable, ensuring the system remains always operational.

PIPELINE FLOW

1. **Product Description**
   Natural language input
2. **Query Expansion**
   Synonym & domain injection
3. **Hybrid Retriever**
   BM25 + Semantic Embeddings
4. **LLM Re-ranking**
   Rationale + context scoring
5. **Top 5 Standards**
   IS number, title & score

System Architecture

Two-phase design: offline indexing + online query processing

OFFLINE PHASE — Build Once

SP-21 PDF Dataset

929 pages · 570+ building material standards across 27 sections

preprocess.py

Regex-based split → 570 structured chunks + section category tags

all-MiniLM-L6-v2 (Embedding Model)

Each chunk encoded as a 384-dimensional dense vector

embeddings.npy + processed_data.json

Persisted to disk — loaded once at startup, zero recompute

ONLINE PHASE — Per Query

User Query (Product Description)

Free-text input via Streamlit UI or inference.py CLI

Query Expansion + LLM Pre-processing

Synonyms injected; LLM reformulates query for broader coverage

BM25 (40%) + Semantic Embedding (60%)

Parallel scoring — min-max normalised then fused by weighted sum

Top 5 BIS Standards Returned

Ranked with IS number, full title, and retrieval confidence score

Stack: Python 3.11 · sentence-transformers (all-MiniLM-L6-v2) · BM25 (custom) · NumPy · Streamlit · pypdf / pdftotext

Chunking & Retrieval Strategy

Every design decision targets accuracy on the hidden private test set

DOCUMENT CHUNKING

1

Regex-based Standard Splitter

Handles 12+ formatting variants (IS:NNN, PART1, year-dash, SEC sub-parts). Each standard becomes exactly one chunk — no partial splits.

2

Section Category Injection

All 27 SP-21 sections (Cement, Steel, Aggregates, etc.) are mapped to keyword tags injected into each chunk, enabling category-aware retrieval.

3

Weighted Text Field for BM25

Enriched text = standard_id×8 + title×12 + category×4 + body. Over-weighting critical fields improves lexical precision significantly.

4

Deduplication of Multi-page Standards

Standards split across PDF pages are merged into a single record before indexing — preventing duplicate or partial matches in retrieval.

RETRIEVAL STRATEGY

BM25 40%

Lexical Keyword Scorer

Tuned with k1=1.4, b=0.70. Reliably handles exact IS number lookups and domain-specific keyword queries.

Embed 60%

Semantic Embeddings (all-MiniLM-L6-v2)

384-dim cosine similarity captures conceptual meaning — finds the correct standard even with zero keyword overlap.

Fusion

Min-Max Normalisation + Weighted Fusion

Both score lists scaled independently to [0,1] before fusion: 0.40×BM25 + 0.60×semantic. Equal scale, tuned weights.

Boost

Explicit IS Number Override

If the query contains a direct IS number (e.g., 'IS 1786'), a +100 score boost ensures rank-1 accuracy and perfect recall.

Demo Highlights

Live at: bis-standard-re-fnf2hb9sywnjf7ptfscutm.streamlit.app

Streamlit web UI · Text area for product description · Returns top-5 ranked recommendations with IS number, title and retrieval score

Query: "ordinary portland cement 33 grade for structural concrete"

Exact grade match via query expansion

→ IS 269 : 1989 — Ordinary Portland Cement, 33 Grade

IS 8112 : 1989 — Ordinary Portland Cement, 43 Grade · IS 455 : 1989 — Portland Slag Cement

Why it works:

Query expansion appends grade synonyms; BM25 locks onto IS number and grade text.

Query: "binder material for masonry walls"

Semantic match — zero shared keywords with standard titles

→ IS 712 : 1984 — Building Limes

IS 2116 : 1980 — Sand for Masonry Mortars · IS 1489 : 1991 — Portland Pozzolana Cement

Why it works:

No keyword overlap with standard titles. Semantic embeddings retrieve by conceptual meaning.

Query: "steel bars for reinforced concrete slabs"

Material category boost + embedding fusion

→ IS 1786 : 2008 — High Strength Deformed Steel Bars

IS 432 — Mild Steel & Medium Tensile Bars · IS 2062 — Hot Rolled Medium & High Tensile Steel

Why it works:

Material category boost amplifies steel-domain chunks; fusion ranks by combined relevance.

Evaluation Results

Public test set (10 queries) — all three automated metric targets exceeded

100%

Hit Rate @3

Target: > 80%

All 10 queries had at least one expected BIS standard in the top-3 recommendations. Perfect recall on the public test set.

1.000

MRR @5

Target: > 0.70

Mean Reciprocal Rank of the first correct standard in top-5 results. Score of 1.0 means the correct standard was always ranked first.

1.30s

Avg Latency

Target: < 5 sec

Average end-to-end response time per query including BM25 scoring, embedding cosine similarity, and result fusion on CPU.

Secondary & Subjective Evaluation Criteria

Relevance Score (Manual, 1–5)

Judges rate top-3 results for 5 random queries. Weighted BM25 text fields and semantic embeddings ensure standard titles and scope closely match the product query.

No Hallucinations (Binary Check)

All returned IS numbers are validated against the indexed SP-21 dataset. The system never fabricates a standard — outputs are drawn strictly from sourced records.

Technical Excellence & Innovation

Hybrid BM25 + semantic fusion, IS-number boost override, category injection, and deduplication represent novel retrieval engineering specific to the BIS domain.

Impact on MSEs

Democratising BIS compliance for 6.3 crore Micro & Small Enterprises in India

BEFORE — Manual Compliance Discovery		AFTER — AI-Powered Recommendation Engine
2–4 Weeks to Identify Standards Employees manually scan hundreds of pages in the SP-21 handbook, often requiring paid legal or technical consultants.	vs	**Results in Under 2 Seconds** A product description typed into the UI or passed to the API returns top-5 ranked BIS standards instantly — no expertise needed.
Requires Specialised Technical Expertise Many MSE owners do not have engineers who understand BIS numbering conventions or category structures.	vs	**Accessible to Any Business Owner** Plain-language descriptions work. The system understands synonyms and trade names through query expansion and semantic search.
High Error Rate — Wrong Standard = Rejection Choosing an outdated or incorrect standard leads to BIS licence rejection, rework costs, and potential legal penalties.	vs	**Rationale Reduces Compliance Risk** Each result shows IS number, full standard title, and retrieval score — helping MSEs verify the match before submission.
Non-Compliance Can Shut Down Business Critical for micro enterprises producing cement, steel, or concrete products where BIS marking is legally mandatory.	vs	**Proactive Compliance Before Product Launch** MSEs can identify and meet the correct standard during development — not after a failed inspection or rejection.

Team & Acknowledgements

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GK

Gufran Khan

Team Lead

Architecture design, hybrid retrieval strategy, fusion logic, and overall project coordination.

UNT

Ujjwal Narayan Thakur

Developer

preprocess.py, retriever.py, BM25 + embedding pipeline, inference.py and evaluation harness.

SK

Subhash Kumar

Developer

Streamlit interface (interface.py), dataset analysis, testing, presentation, and documentation.

Built with:

- Python 3.11
- sentence-transformers
- BM25 (custom)
- Streamlit
- pypdf / pdftotext
- NumPy

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